ETHICAL HACKING PRACTICAL FILE

Practicals 2, 3, 7, 9 and 10

Short practical notes with simple explanations. No sample outputs or screenshot pages are included.

Practical 2: RC4 Encryption and Decryption Using CrypTool

Aim: To encrypt and decrypt sample text using RC4 and understand sample hash calculation.

Requirements: CrypTool 1 and a Windows computer. Cain & Abel is optional for the hash-calculator demonstration.

Theory: RC4 is a symmetric stream cipher. The same secret key is used to encrypt and decrypt data. MD5 creates a fixed-length hash; it is not encryption and cannot directly be reversed. RC4 and MD5 are outdated and are used here only for learning.

Procedure

1. Open CrypTool and create a new text document.
2. Type a sample message such as HELLO WORLD and select it.
3. Choose Encrypt/Decrypt → Symmetric (modern) → RC4.
4. Enter a temporary test key in the required format and select Encrypt. The text is converted into ciphertext.
5. Select the ciphertext, open RC4 again, enter the same key and choose Decrypt.
6. Check that the original sample message is recovered.
7. For a hash demonstration, calculate the MD5 digest of a self-created sample word using a hash calculator or the Python hashlib library.

Explanation: Encryption changes readable plaintext into ciphertext. Decryption with the same key restores the plaintext. A hash is a one-way digest used for integrity and password-verification concepts.

Precautions: Use only sample text and a temporary key. Never test real account passwords or another person's credentials.

Practical 3: Linux Networking Commands and ARP Poisoning Concept

Aim: To study basic Linux networking commands and understand the security risk of ARP poisoning.

Requirements: Linux terminal or Ubuntu on WSL; Cain & Abel may be used only for instructor-supervised discussion in an isolated lab.

Theory: Networking commands display interface details, connections, reachability and packet routes. ARP maps an IPv4 address to a MAC address on a local network. ARP poisoning means supplying false ARP information, which may redirect local traffic.

Procedure

ifconfig — Displays network interfaces, IP addresses and MAC addresses. If unavailable, use `ip addr`.

netstat -an — Displays active connections and listening ports. If unavailable, use `ss -an`.

ping -c 4 8.8.8.8 — Sends four test packets to check network reachability.

traceroute 8.8.8.8 — Displays the network hops between the computer and destination. If unavailable, use `tracepath` where installed.

8. Open the Linux terminal or Ubuntu on WSL.
9. Run each command above and read the information it displays.
10. To understand ARP, view the local ARP cache using `ip neigh` (Linux) or `arp -a` (Windows).

11. Discuss how a false IP-to-MAC mapping could redirect traffic. Do not perform poisoning on a live or shared network.

Precautions: Use your own lab devices only. Do not target college, public Wi-Fi or other users. Any live demonstration must be isolated and supervised.

Practical 7: Data Tampering Using Firefox, FoxyProxy and Burp Suite

Aim: To inspect an HTTP request and understand controlled modification of a harmless form value in a local test application.

Requirements: Firefox, FoxyProxy Standard, Burp Suite Community Edition and a local practice application such as DVWA.

Theory: A browser sends HTTP requests to a server. Burp Suite acts as an intercepting proxy and allows a tester to inspect requests. FoxyProxy routes browser traffic through the proxy.

Procedure

12. Start the local test application and open it in Firefox using its localhost address.
13. Open Burp Suite and start its Proxy listener. Confirm the listener address and port; the common default is 127.0.0.1:8080.
14. In FoxyProxy, create a profile with the same host and port as the Burp listener, then enable that profile.
15. In Burp Suite, open Proxy → Intercept and turn interception on.
16. Submit a harmless form in the local test application using dummy information.
17. Inspect the intercepted request and identify a non-sensitive test parameter.
18. Change only that dummy parameter and click Forward. Observe how the local application handles the changed value.
19. Turn interception off and disable the FoxyProxy profile when finished.

Explanation: The proxy displays the request before it reaches the server. Changing a test parameter demonstrates that data sent by a browser can be modified unless the application validates it.

Precautions: Use localhost and dummy values only. Never intercept real credentials, cookies, tokens, purchases or third-party websites.

Practical 9: Python GUI Text Logger Using Tkinter

Aim: To create a visible GUI that saves text deliberately entered by the user.

Requirements: Python 3 and Tkinter.

Theory: Tkinter is Python's standard GUI library. The program below reads text only from its own text box when the Save button is clicked. It does not monitor keyboard activity in other applications.

Program

```
import tkinter as tk

def save():
    text = box.get("1.0", "end-1c")
    with open("key_log.txt", "a") as f:
        f.write(text + "\\n")
    status.config(text="Text saved successfully")

root = tk.Tk()
root.title("Text Logger")

box = tk.Text(root, height=8, width=40)
```

```
box.pack()

tk.Button(root, text="Save", command=save).pack()

status = tk.Label(root, text="")
status.pack()

root.mainloop()
```

Procedure

20. Open VS Code or IDLE and create a file named `text_logger.py`.
21. Type the program and save the file.
22. Run it using `python text_logger.py`.
23. Enter your own sample text in the text box and click Save.
24. The program appends the entered text to `key_log.txt` in the working folder.

Explanation: `box.get()` reads the text area. The file is opened in append mode (`a`), so new entries are added without deleting earlier entries. The Save button runs the `save()` function.

Precautions: Record only text knowingly entered by the user. Do not add hidden keyboard monitoring or collect passwords or private information.

Practical 10: Metasploit Framework – Windows Lab Assessment

Aim: To explore Metasploit and perform a limited TCP port scan against an authorised Windows lab machine.

Requirements: Kali Linux with Metasploit Framework, a Windows test VM and VirtualBox or VMware.

Theory: Metasploit is a penetration-testing framework. Auxiliary scanner modules can help identify reachable ports on an authorised target. A port scan does not itself prove that a service is vulnerable.

Safe Lab Setup

25. Use Kali and Windows virtual machines on the same Host-only or Internal Network.
26. Keep Windows Defender and the firewall enabled. Take a VM snapshot before testing.
27. Find the Windows lab VM's IP address with `ipconfig`. Use only this lab IP.

Procedure

28. Start Kali Linux and open Terminal.
29. Launch Metasploit by entering `msfconsole`.
30. Search for the TCP port scanner using `search portscan`.
31. Select the scanner with `use auxiliary/scanner/portscan/tcp`.
32. View its settings using `show options`.
33. Set the authorised Windows lab IP using `set RHOSTS <Windows-Lab-IP>`.
34. Set a limited port range using `set PORTS 1-1024`.
35. Start the scan using `run` and review the ports reported by the scanner.
36. Exit Metasploit using `exit` and restore the lab environment.

Important Commands

```
msfconsole\nsearch portscan\nuse auxiliary/scanner/portscan/tcp\nshow options\nset RHOSTS <Windows-Lab-IP>\nset PORTS 1-1024\nrun\nexit
```

Explanation: `RHOSTS` specifies the authorised target. `PORTS` limits the range being checked. The scanner reports ports that respond; results depend on firewall rules and running services.

Precautions: Scan only your own isolated lab machine or a system with written permission. Do not create or execute payloads, disable security software or scan public/college networks.